11 DETERMINE THE POROSITY FOR THE GIVEN SAND SAMPLE

Aim

To determine the Porosity for the given sand sample.

Definition

Porosity of the sand is the ratio of volume of voids to the volume of given sand mass. It is mentioned in percentage.

$$\text{Porosity 'n'} = \frac{\text{volume of voids}}{\text{Total volume of sand}} \times 100$$

TJIPPQ

Apparatus and Materials Required

1. Weighing Balance
2. Bucket
3. Sand
4. Water
5. 16mm dia, 600mm length tamping rod

Procedure

1. Take a clean dry bucket and weigh it (W_1 grams).
2. Take about 1/3rd bucket of clean sand and tamp it 25 times. Do the same twice to fill the bucket.
3. Weigh the bucket with sand (W_2 grams).
4. Now fill the bucket with sand, with water and weigh it (W_3 grams).
5. Now empty the bucket, fill it with water and weigh it (W_4 grams).

Observation

1. Weight of the empty bucket (W_1 grams). =
2. Weight of the bucket + sand (W_2 grams). =
3. Weight of the bucket + sand + water (W_3 grams). =
4. Weight of the bucket + water (W_4 grams). =

$$\text{Porosity 'n'} = \frac{(W_3 - W_2)}{(W_4 - W_1)} \times 100 \quad =$$

Where

V_v = Volume of voids
V_s = Volume of solids
V_g = Volume of air
V_w = Volume of water

Result

Porosity of the given sand sample is ____%

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